

Dataset Collection Protocol

How to record, what to record, and how much — hand this to anyone collecting data

Data collection is **calendar-bound, not effort-bound** — it cannot be compressed into a weekend before the deadline. Start immediately and run it in parallel with hardware and model work. This document is written so a teammate can run a full recording session without asking anyone questions.

60+ speakers total (minimum 40)	~70 utterances per speaker	~4,200 real positives before augmentation	~12 min session length per speaker
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If forced to choose, prioritise speaker count over per-speaker depth. 60 speakers × 20 utterances generalises better than 20 speakers × 60. A model is only as good as the variety of vocal tracts it has heard.

Two design rules that must not be broken

1. Every speaker records every condition — never one condition per person

If speaker 1 does normal voice and speaker 2 does the noisy condition, then “noisy” and “speaker 2’s voice” become perfectly correlated and the model cannot separate them. It learns the wrong thing, tests beautifully, and collapses on an unseen speaker at the finale. The design is **crossed**, not split: every person walks through the whole script below.

2. Negatives must come from the same speakers, phones and rooms as the positives

If all positives are your classmates on phones and all negatives come from MUSAN or Speech Commands, the model takes a shortcut: it detects *the recording setup*, not the word “Kunjika”. Accuracy will look superb and false activations at the finale will be catastrophic. This is why the script below includes confusable words and free speech from every single speaker.

Record humans; simulate everything else

Human recording time is the scarcest resource in this project. Spend it only on things that cannot be generated by the augmentation pipeline.

Do NOT record — simulate it	Why	MUST record — cannot be simulated
Distance from mic, room reverb	RIR convolution gives exact, controllable reverb across hundreds of virtual rooms	Speaker identity — vocal tract, pitch range, timbre
Background chatter, TV, fans, traffic	Noise mixing from MUSAN / DEMAND at precisely controlled SNR levels	Accent and mother tongue — get Hindi, Punjabi, Bengali, Tamil, Malayalam, Marathi speakers
Someone talking in front while user speaks from behind	This is just additive mixing — we can generate 10,000 versions of it	Pronunciation variants — KUN-ji-ka vs kun-JI-ka, hard vs soft <i>j</i>
Volume / gain variation, mic frequency response	Gain scaling and channel filtering, applied in software	Lombard speech — people genuinely change pitch and spectral tilt in noise. Adding noise to a quiet clip does <i>not</i> reproduce this.

This is why the script has only **one** genuinely noisy-environment condition. Everything else is recorded clean and close, then multiplied in software.

Why 30 repeats and not 2

Say the word ten times and the clips will differ measurably: pitch wanders by 10–20 Hz, the *k* burst energy varies, vowel formants shift, the *j* gets more or less affricated, duration ranges roughly 550–750 ms. That is not a condition anyone chose — it is the natural noise floor of human articulation, and it is exactly what the model must learn to ignore.

Two examples teach the model that Kunjika sounds like *those two specific waveforms*. Thirty teach it the acoustic *region* the word occupies. The condition folders (fast, slow, quiet, loud) sample **deliberate axes**; the 30 normal repeats sample the **random variation inside any one axis**. Both are needed.

Augmentation multiplies variety — it does not create it

Run 2 clips through reverb and noise mixing and you get 2,000 files, but they are 2,000 variations of *two* articulations. Every copy inherits the same formant trajectory and the same duration. The model ends up robust to rooms and brittle to people. Multiply a small number and it is still a small number.

Also budget for loss: realistically 10–20% of clips are unusable (clipped, truncated, a cough, wrong word, phone glitch). Recording 30 to keep 25 is normal and expected.

The recording script — run this with every speaker

Roughly 12 minutes per person. Work top to bottom; do not skip ahead.

#	Folder	Reps	What to say	How to say it
1	normal	30	Kunjika	Seated, arm's length (~30 cm), normal conversational voice, brief pause between each
2	fast	10	Kunjika	Noticeably rushed, as if in a hurry
3	slow	10	Kunjika	Deliberately drawn out, over-enunciated
4	quiet	10	Kunjika	Soft, near-whisper but still voiced — not a true whisper
5	loud	10	Kunjika	Raised voice, as if calling from across a room
6	lombard	10	Kunjika	Move to a genuinely noisy place — corridor, canteen, near traffic. Speak naturally over the noise. This captures the real Lombard effect and cannot be faked in software.
7	negatives	20 total	kunji ka / kunji / kunj / kanjika / punjika / anjika	About 3–4 reps of each, normal voice. These are the hard negatives — the most valuable recordings in the whole session.
8	freesepeech	90 sec	Ordinary conversation, Hindi and English	Describe your day, your project, anything. One continuous take. Must not contain the word Kunjika.
9	silence	5 sec	— nothing —	Room tone with the speaker present but silent. One take.

Why “kunji ka” is the single most important negative

Kunji ka is a natural Hindi phrase meaning “of the key”, and it **will** occur in ordinary speech near the device. If it is not in the hard-negative set, we get a mysterious false-activation problem during a Hindi-speaking judge's demo. Never skip step 7.

Recording settings

Record on phones, not on the INMP441. We cannot queue 60 people at one breadboard, and speaker count is what drives generalisation. But phone audio must be captured correctly or it is useless.

Setting	Required value	Why it matters
Format	WAV, 16-bit PCM — never MP3, M4A or AAC	Lossy codecs discard exactly the high-frequency detail the <i>k</i> burst and <i>j</i> affricate live in
Sample rate	16 kHz minimum; 44.1 or 48 kHz is fine and we downsample	Our pipeline runs at 16 kHz. Higher is safe; lower is unrecoverable
Channels	Mono	The device has one mic

Setting	Required value	Why it matters
Noise suppression	OFF where the app allows it	Phone NS is aggressive and non-linear; it removes cues the model needs
Automatic gain (AGC)	OFF where the app allows it	AGC destroys the loud/quiet distinction we are deliberately recording in steps 4 and 5
Android app	RecForge II, or Audio Recorder (open source)	Both expose PCM WAV and let you disable processing
iPhone app	Any WAV recorder — not Voice Memos	Voice Memos outputs compressed AAC only

Session hygiene

- **One utterance per button press.** Do not record a long continuous take and segment it later — segmenting is tedious, error-prone, and costs a full day. Capturing individually costs nothing.
- **Hold the phone at roughly 30 cm**, not against the mouth. Close-mic recording adds plosive pops and an unnatural bass boost we will never see on the real device.
- **Do not record in a moving vehicle or in wind.** Wind noise cannot be augmented away.
- **Listen back to the first three clips** before continuing the session. Catching a bad phone setting at clip 3 saves the whole session; catching it at clip 70 does not.
- **Same phone for the whole session.** If a speaker switches devices halfway, note it in the metadata.

File naming and folder structure

Organise **speaker-first, condition second**. We will be doing speaker-disjoint train/test splits constantly, and this structure makes that a one-line glob instead of a filtering headache.

```
kunjika_data/
  spk_007/
    normal/    spk_007_normal_01.wav ... _30.wav
    fast/      spk_007_fast_01.wav   ... _10.wav
    slow/      spk_007_slow_01.wav
    quiet/     spk_007_quiet_01.wav
    loud/      spk_007_loud_01.wav
    lombard/   spk_007_lombard_01.wav
    negatives/ spk_007_neg_kunjika_01.wav
              spk_007_neg_kanjika_01.wav
    freespeech/ spk_007_free_01.wav
    silence/   spk_007_silence_01.wav
  spk_008/ ...
  metadata.csv
```

Use **speaker IDs, not names**. It keeps the consent story simple and lets us publish figures without exposing anyone. Keep the ID→name mapping in one private sheet held by the team lead only.

metadata.csv — one row per speaker

```
speaker_id,age_band,gender,mother_tongue,phone_model,room,date,notes
spk_007,18-22,F,Bengali,Redmi Note 12,hostel_room,2026-09-09,
spk_008,18-22,M,Tamil,iPhone 13,library,2026-09-09,mild cold
```

The **mother_tongue** and **phone_model** columns are not bureaucracy — they let us check for hidden correlations later (e.g. all Tamil speakers happened to use one phone model) and they let us report speaker diversity in the deck, which judges notice.

Who to recruit

ISRO is a national organisation and the finale judges will not all sound like Delhi. Accent diversity is the highest-value axis we can control. Rough targets across the full 60:

Axis	Target spread	Note
Mother tongue	At least 5 distinct languages; no single one above ~40%	Hindi, Punjabi, Bengali, Tamil, Telugu, Malayalam, Marathi — whatever the campus offers
Gender	Roughly balanced, minimum 35% either way	Pitch range is one of the strongest acoustic axes; a skewed set fails on half the judges
Age	Mostly 18–25, but recruit 5–10 speakers over 35	Faculty, staff, canteen workers, family. Older voices differ measurably and will otherwise be absent
Phone model	As many distinct models as possible	Channel diversity for free — different mics, different processing
Room	Vary it — hostel, classroom, lab, home, corridor	Cheap real reverb diversity on top of what we simulate

The device-matched calibration set

Phone mics apply processing, roll off differently, and sit behind different acoustics than the INMP441. A model trained purely on phone audio measurably degrades on raw device PCM. This is called **channel mismatch** and it is handled in three parts:

Step	When	What to do
Record clean phone corpus	Now — days 0–14	The 60-speaker protocol above. This is the bulk of the training data.
Record device-matched set	Once the ESP32-S3 + INMP441 is capturing audio (~day 5)	All 6 team members plus anyone nearby record 40 utterances each through the INMP441 . A few hundred clips. Not enough to train on — this is the threshold-calibration and final validation set. It is where we find out whether the phone-trained model actually transfers.
Apply channel augmentation	Once both sets exist	Record the same audio through a phone and through the INMP441, derive the response difference, and apply it as a filter across the phone corpus. Cheap and effective.

Quality control — do this weekly, not at the end

Check	How	Action if it fails
Clipping	Peak amplitude at or above full scale	Discard the clip; re-record the speaker at lower gain if widespread
Truncation	Word cut off at the start or end of the file	Discard. Usually means the speaker started before the button; re-brief them
Wrong word	Spot-listen 5 random clips per speaker	Discard the clip. If several, re-run the session
Silence / dead file	File duration under ~0.3 s or near-zero RMS	Discard. Often a phone permissions issue — check the whole session
Compressed format	File is .m4a / .mp3, or WAV header shows a codec	Re-record the whole session. This cannot be repaired
Kunjika inside freespeech	Listen to the 90 s take	Trim that segment out, or drop the file — a positive hiding in the negatives poisons training

Write a **10-line Python script** on day 1 that walks the folder tree and flags duration outliers, clipping and near-silence automatically. Running it after every batch of speakers turns quality control from a two-day panic into a two-minute habit.

The split rule — get this wrong and nothing else matters

Split by speaker, never by clip

If speaker 07 appears in both the training and the test set, the model can recognise *that person's voice* rather than the word, and the reported accuracy is inflated — sometimes by 15 points or more. We will not find out until the finale, in front of judges. **Hold out 15 speakers entirely**, chosen to span the accent and gender range, and do not look at them until threshold calibration.

Split	Speakers	Used for
Train	~35 speakers	Model fitting, all augmentation applied here
Validation	~10 speakers	Architecture choices, early stopping, hyperparameters
Test (held out)	~15 speakers	Touched once, at the end. Threshold calibration and reported numbers

Consent

Keep it light but do it — judges may ask, and it costs nothing. Read this aloud or put it at the top of the sign-up form:

I agree to my voice being recorded for the SIH 2026 project team's keyword-spotting model. Recordings will be stored under an anonymous speaker ID, used only to train and evaluate this model, and not shared publicly or used for any other purpose. I can ask for my recordings to be deleted at any time.

Log a yes/no consent column in metadata.csv. If someone later asks for deletion, the speaker-first folder structure makes it a single `rm -r spk_0NN`.

Common mistakes, in order of how much damage they cause

Mistake	Consequence
One condition assigned per person	Condition and speaker become inseparable; the model learns the wrong feature
Negatives sourced only from public datasets	Model detects our recording setup, not the word. Catastrophic false activations
Splitting train/test by clip instead of speaker	Reported accuracy inflated by 10–20 points; discovered at the finale
Recording in MP3 / M4A / AAC	Unrecoverable. The entire session must be re-recorded
Only 1–2 repeats per speaker	Augmentation multiplies two articulations; model is brittle to new speakers
Recording distance and background noise by hand	Wasted human time on things software generates better and in greater volume
Leaving quality control until the end	Discovering a broken phone setting after 40 sessions instead of after 1
Storing names instead of speaker IDs	Awkward consent story and messy deletion requests

Questions during a session go to the team lead, not to guesswork — a consistent wrong choice is recoverable, an inconsistent one is not.